

# Fake News Detection with Machine Learning

IBM PBEL Virtual Internship 2026 — Artificial Intelligence

Presented by

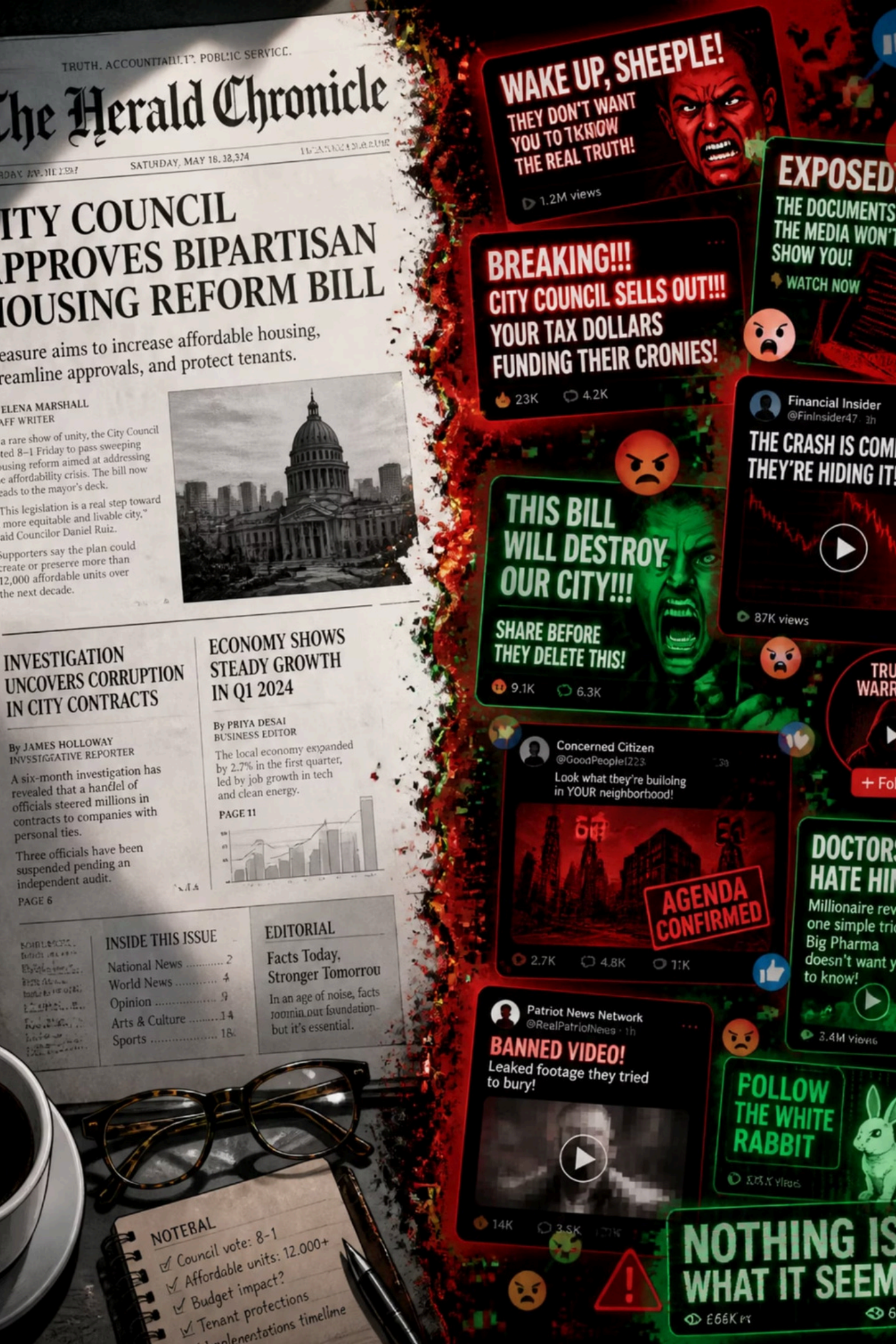
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# Introduction

## What Is Fake News?

False or misleading content passed off as real news.

## Why It Matters

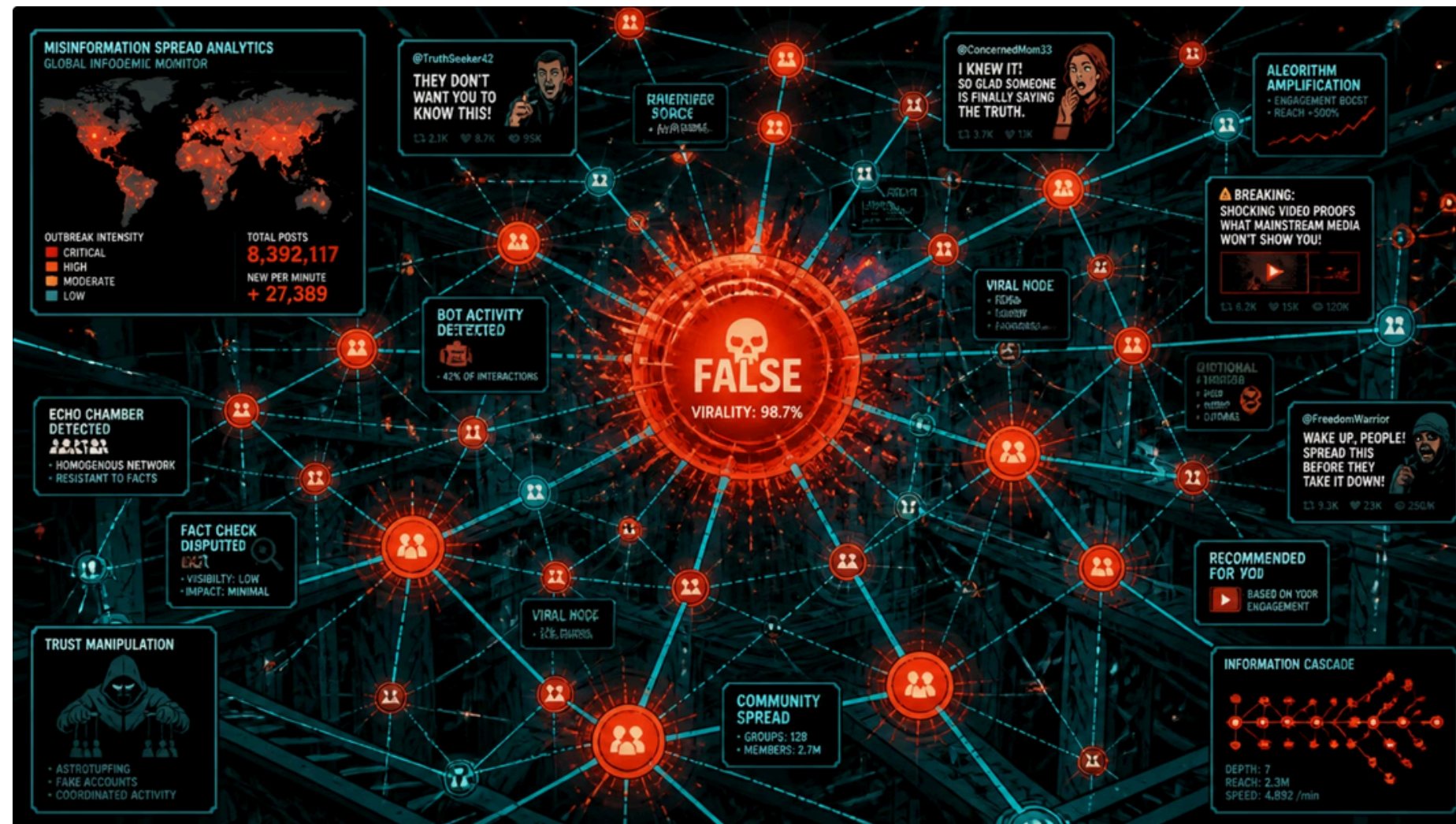
It spreads fast on social media and shapes opinions and elections.

## Our Approach

We use ML and NLP to classify articles as **Real** or **Fake**.



# The Challenge



## ⚡ Fast Spread

False claims reach millions in minutes

## 🔍 Manual Checks

Too slow to scale

## 🌐 Real Impact

It shapes elections, markets, and trust

## ✅ Why It Matters

We need accurate automated detection

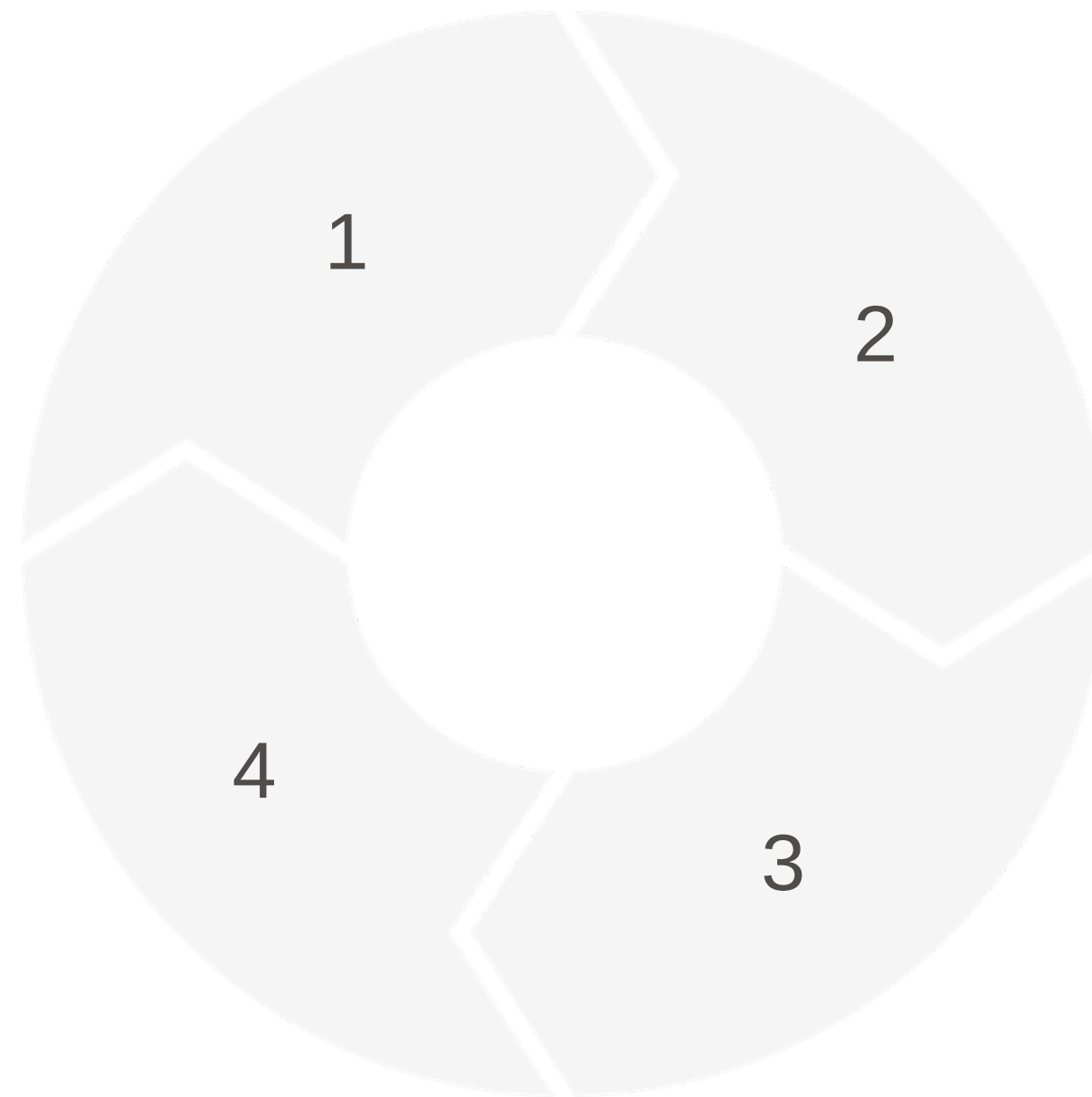
# Project Objectives

## Fake News Detection

Detect fake news with machine learning and NLP.

## Performance Evaluation

Measure results with standard metrics.



## TF-IDF Features

Convert text into numbers.

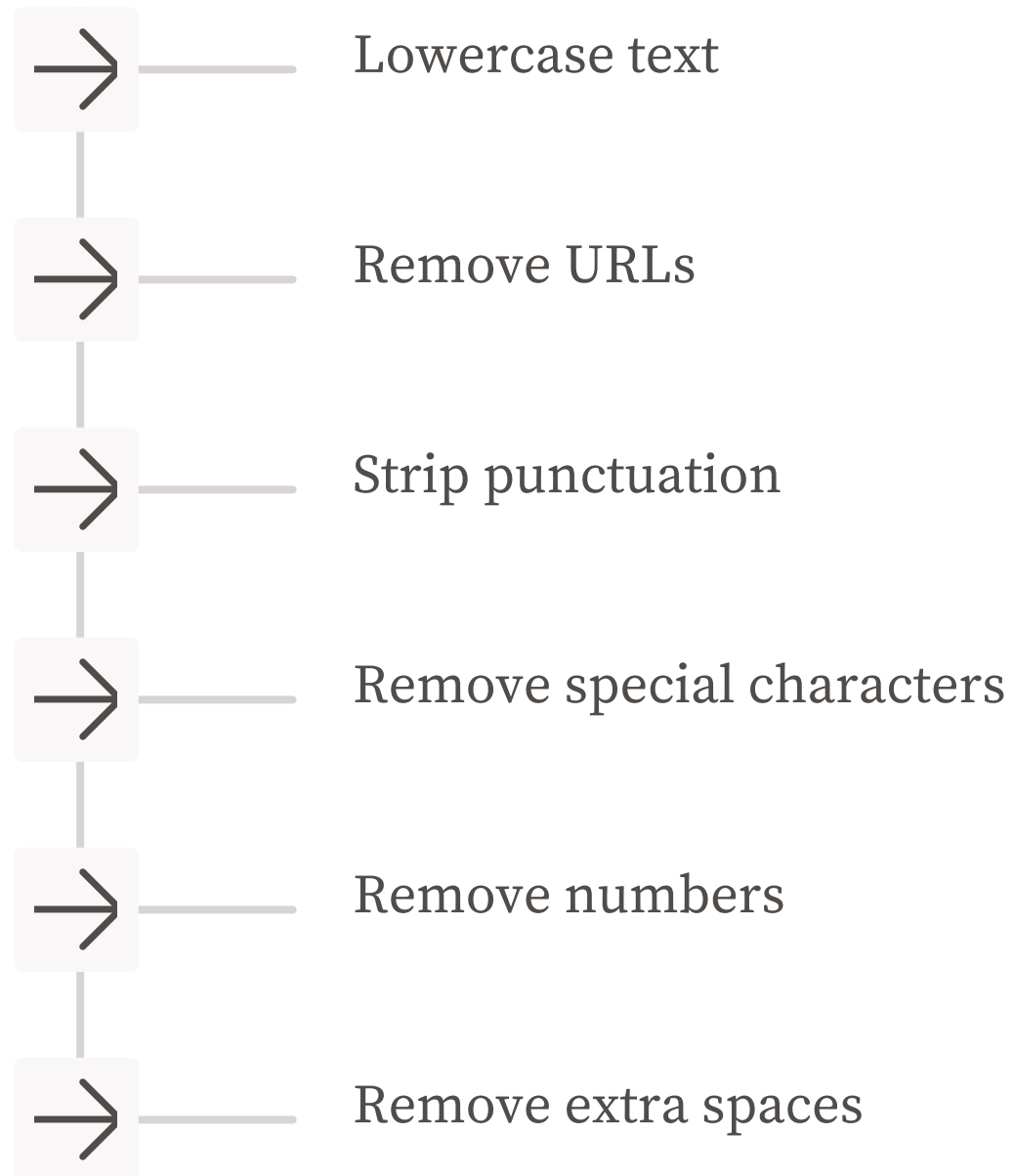
## Compare Classifiers

Test multiple models.

# Data Preprocessing

## Text Cleaning Steps

This sequence prepares raw text for modeling.



# Technologies Used



Python

Core language



Pandas

Load and clean data



NumPy

Fast numerical computing



NLTK

Text processing



TF-IDF

Text feature extraction



Matplotlib

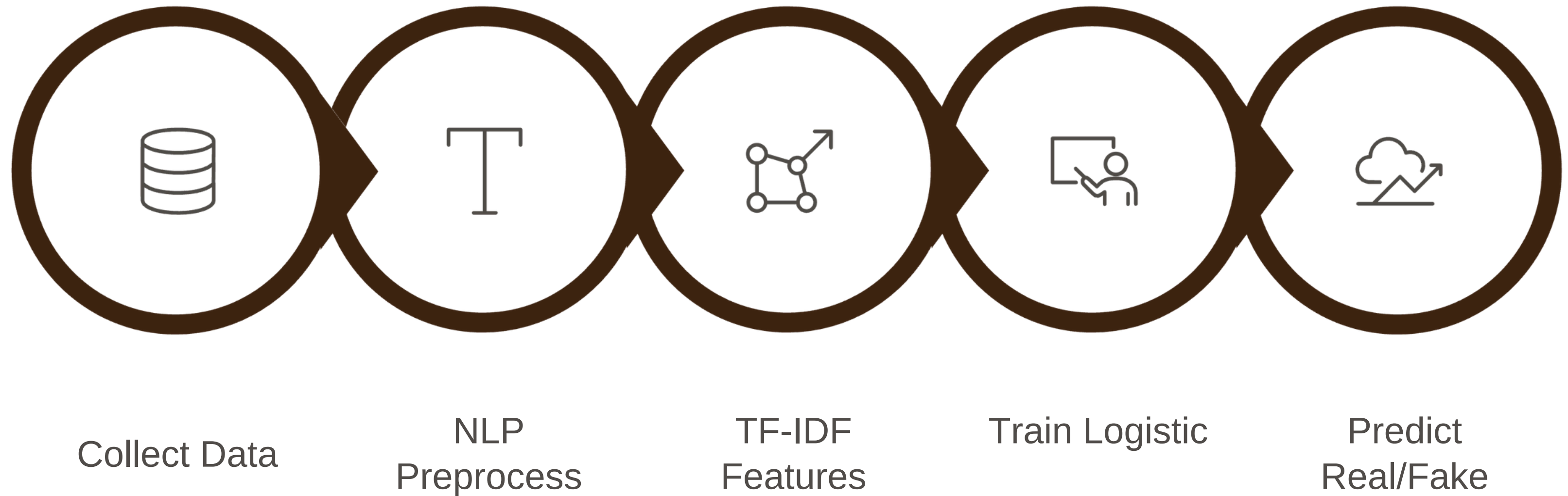
Data visualization



Jupyter

Build and test notebooks

# Methodology



Each stage turns raw text into a clear classification signal — collect, clean, vectorize, train, predict.



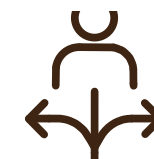
# Machine Learning Models



## Logistic Regression

Fast and effective

Strong for text classification



## Decision Tree

Rule-based

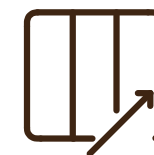
Easy to interpret



## Random Forest

Ensemble method

Helps reduce overfitting



## Gradient Boosting

Builds sequentially

High accuracy



# Results & Performance

## Accuracy

**90%** — 90 correct predictions out of 100 total

## Precision

**87.5%** — 35 true positives out of 40 predicted fakes

## Recall

**90%** — 45 true positives out of 50 actual fakes

## F1-Score

**84.7%** — harmonic mean of precision and recall

📌 TF-IDF improved performance on unseen articles.

Thank You